

GEORGE CHEMMALA

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EDUCATION

Brown University

Providence, RI

Sc.B. Computer Science, Sc.B. Mathematics (Major GPA: 4.0)

May 2026

- Machine Learning | Deep Learning | Systems | Compilers | Programming Languages | Robust Algorithms for ML | Formal Proof and Verification | Applied & Pure Cryptography | Abstract Algebra | Number Theory | AI & Euclidean Geometry

EXPERIENCE

Software Engineer Intern, Full Stack | Aiguard

Summer 2025

- Deployed a scalable full-stack web platform for solar installation cancellation prediction and analysis, integrating FastAPI and PostgreSQL for robust backend APIs working with GCP and React/TypeScript for a responsive frontend
- Developed an efficient Boosted Random Forest Classifier (86% accuracy) and Forecaster for actionable insights for the client
- Added model interpretability using SHAP/LIME (AI explanation tools), building visual dashboards to show stakeholders the top drivers of solar installation cancellations
- Refactored backend codebase for modularity and extensibility, applying OOP principles and reducing technical debt, resulting in a 2x reduction in code size and improved maintainability

Research Assistant | Brown Department of Computer Science & Physics

Summer 2025 -Present

- Implemented a novel transformer pruning method on the ParT particle jet classifier to be used on FPGAs at CERN, resulting in an expected 30% reduction in model size and improved inference speed
- Surveyed state-of-the-art transformer pruning techniques and utilized advanced PyTorch features, including model hooks, to design and integrate custom pruning logic that utilized GPU support
- Integrated the pruning pipeline into the existing workflow using the Weaver framework, ensuring reproducibility

President | Math Circle

Spring 2023 - Present

- Head a student-run outreach program teaching recreational math lessons (e.g., hexaflexagons, SET, combinatorial games) to local high school and middle school students (5th-12th grade)

Teaching Assistant | Department of Mathematics & Applied Mathematics & Computer Science

Spring 2024 - Present

- Head TA for Machine Learning (Spring 2026), Theory of Computation, Applied ODEs, Stats Inference, Abstract Algebra
- Led course logistics, coordinated TA team, spearheaded problem-solving sessions, held office hours, and graded advanced coursework (including proof-based reasoning) for classes with up to 350 students

Computer Technician and Teaching Assistant | Georgia's Governor's Honors Program (GHP)

Summer 2023

- Coordinated instructional support (intro programming and deep learning) and mentored students on computational mathematics research projects for a cohort of 80 students across multiple classrooms (each with ~30 students)
- Assisted students with debugging Python code, troubleshooting hardware/software issues, and optimizing ML workflows

PROJECTS/RESEARCH

Vote: Cryptographic Voting Protocol Implementation | C++, CryptoPP

Spring 2025

- Built a secure, end-to-end verifiable voting system inspired by Helios, using homomorphic encryption and threshold decryption for distributed trust among 2+ arbiters
- Integrated zero-knowledge proofs and blinding to protect voter anonymity and ensure ballot integrity
- Extended the protocol to support multiple candidates and a restriction on the number of votes per user using a ZKP

WeensyOS: Virtual Memory Kernel Implementation | C, C++, x86 Assembly

Spring 2025

- Engineered kernel code to manage user/kernel virtual memory and enforce correct access permissions
- Enhanced memory utilization by 3x, dynamically allocating virtual pages to non-contiguous physical memory
- Optimized fork and exit system calls for efficient process creation/exit with shared memory
- Incorporated process isolation by overlapping virtual address spaces, enabling full process virtual space utilization

Robust Estimation for the Erdős-Rényi Model | Python, Matplotlib, NumPy, NetworkX

Fall 2024

- Developed and implemented robust algorithms to estimate the edge probability p in Erdős-Rényi graphs under adversarial edge perturbations for adversarial perturbation $\varepsilon < 1/2$
- Proved partial theoretical guarantees on runtime and estimation accuracy, outperforming existing methods
- Analyzed empirical evaluations on synthetic graph data (+10k nodes) on department servers, demonstrating improvements in both computational efficiency and statistical robustness
- Applied software engineering best practices including modular design, visualization with Matplotlib, and reproducibility via clear documentation

SKILLS & INTERESTS

Languages: Python, C, C++, OCaml, Java, JavaScript, Lean, Isabelle, Haskell, Bash, Julia, LaTeX, Mathematica, MATLAB

Developer Tools: VS Code, Vim, PyCharm, IntelliJ, Google Colab, Jupyter, Git, Unix, Linux, Docker, Figma

Libraries: pandas, NumPy, scikit-learn, Catboost, Matplotlib, NetworkX, TensorFlow, PyTorch, TorchVision, OpenCV, CUDA

Interests: Education, Music, Board/Card Games, Photography, Manim